Exercises

Exercise 3.1 · Exponential forms. Write in exponential form: $-2+2i$; $-1-i\sqrt{3}$; $5i$; $\dfrac{1+i}{1-i}$.

Exercise 3.2 · Two ways of writing the same product. Let $z_1 = \sqrt{3}+i$ and $z_2 = 1-i$.

1. Write z_1, z_2 and then $z_1 z_2$ in exponential form.
2. Write $z_1 z_2$ in algebraic form.
3. Deduce the exact values of $\cos\frac{\pi}{12}$ and $\sin\frac{\pi}{12}$.

Exercise 3.3 · Powers.

1. Compute $(1-i)^8$ and $\left(\sqrt{3}+i\right)^{12}$.
2. Determine the natural numbers $n \geq 1$ for which $\left(\sqrt{3}+i\right)^n$ is real; positive real; purely imaginary. Give the smallest in each case.

Exercise 3.4 · De Moivre's formula. Using de Moivre's formula and the binomial formula, express $\cos 4\theta$ in terms of $\cos\theta$ only, and $\sin 4\theta$ in terms of $\cos\theta$ and $\sin\theta$.

Exercise 3.5 · Linearising in order to integrate.

1. Using Euler's formulas, prove that $\sin^3 x = \dfrac{3\sin x - \sin 3x}{4}$.
2. Deduce the value of $\displaystyle\int_0^{\pi} \sin^3 x \,\mathrm{d}x$.

Exercise 3.6 · Modulus and argument of $1+\mathrm{e}^{i\theta}$.

1. Prove that for every real number θ, $1+\mathrm{e}^{i\theta} = 2\cos\frac{\theta}{2}\,\mathrm{e}^{i\theta/2}$.
2. Deduce the modulus and an argument of $1+\mathrm{e}^{i\theta}$ when $\theta \in (-\pi, \pi)$.
3. Application: modulus and argument of $1+\mathrm{e}^{2i\pi/3}$, then a check with the algebraic form.

Exercise 3.7 · A sum of cosines. Let θ be a real number that is not a multiple of 2π, and n a natural number.

1. Prove that $\displaystyle\sum_{k=0}^{n} \mathrm{e}^{ik\theta} = \frac{\mathrm{e}^{i(n+1)\theta}-1}{\mathrm{e}^{i\theta}-1}$.
2. Using the half-angle method, prove that this sum equals $\mathrm{e}^{in\theta/2}\dfrac{\sin\frac{(n+1)\theta}{2}}{\sin\frac{\theta}{2}}$.
3. Deduce $\displaystyle\sum_{k=0}^{n} \cos(k\theta) = \frac{\cos\frac{n\theta}{2}\,\sin\frac{(n+1)\theta}{2}}{\sin\frac{\theta}{2}}$, then check the formula for $n=2$ and $\theta=\frac{\pi}{2}$.